

Master Memory Manager Deployment Guide

Overview

This guide provides comprehensive instructions for deploying the Master Memory Manager (MMM) Phase 4 in production environments.

Prerequisites

System Requirements

- **CPU:** x86-64 architecture, minimum 4 cores, recommended 8+ cores
- **Memory:** Minimum 8GB RAM, recommended 16GB+ for production
- **Storage:** Minimum 10GB free space, SSD recommended
- **OS:** Linux kernel 5.4+, Ubuntu 20.04+ or equivalent

Dependencies

```
# Core dependencies
sudo apt-get update
sudo apt-get install -y build-essential cmake gcc-multilib
sudo apt-get install -y libc6-dev linux-headers-$(uname -r)

# Performance monitoring dependencies
sudo apt-get install -y perf-tools-unstable htop iotop

# Security dependencies
sudo apt-get install -y libssl-dev libcrypt-dev

# AI/ML dependencies (optional)
sudo apt-get install -y python3-dev python3-pip
pip3 install numpy scipy scikit-learn
```

Build Process

1. Clone and Setup

```
# Clone repository
git clone https://github.com/Mecca-Research/The-Binary-Decomposition-Interface.git
cd The-Binary-Decomposition-Interface

# Switch to production branch
git checkout main

# Create build directory
mkdir -p build && cd build
```

2. Configure Build

```
# Configure with CMake
cmake .. -DCMAKE_BUILD_TYPE=Release \
        -DMMM_ENABLE_AI=ON \
        -DMMM_ENABLE_SECURITY=ON \
        -DMMM_ENABLE_TELEMETRY=ON \
        -DMMM_PRODUCTION_BUILD=ON

# Alternative: Debug build for development
cmake .. -DCMAKE_BUILD_TYPE=Debug \
        -DMMM_ENABLE_TESTING=ON \
        -DMMM_ENABLE_PROFILING=ON
```

3. Build System

```
# Build with parallel jobs
make -j$(nproc)

# Run tests (optional but recommended)
make test

# Install system-wide (requires sudo)
sudo make install
```

Configuration

1. System Configuration

Create `/etc/mmm/mmm.conf` :

```

[system]
# Core system settings
max_memory_pools = 64
default_pool_size = 1048576
optimization_level = 3
thread_pool_size = 8

[performance]
# Performance tuning
cache_optimization = true
adaptive_optimization = true
performance_monitoring = true
telemetry_interval = 1000

[ai]
# AI capabilities
enable_predictive = true
enable_anomaly_detection = true
enable_self_healing = true
learning_rate = 0.01
prediction_window = 10000

[security]
# Security settings
encryption_level = 256
audit_logging = true
access_control = strict
security_policy = /etc/mmm/security.policy

[monitoring]
# Monitoring and telemetry
enable_telemetry = true
telemetry_endpoint = http://localhost:8080/telemetry
alert_threshold_memory = 90
alert_threshold_cpu = 85

```

2. Security Configuration

Create `/etc/mmm/security.policy` :

```

# MMM Security Policy
# Access control rules
allow user:root operation:*
allow group:mmm-admin operation:configure,monitor
allow group:mmm-user operation:allocate,free

# Encryption settings
encrypt_memory_pools = true
encrypt_telemetry = true
key_rotation_interval = 86400

# Audit settings
audit_all_operations = true
audit_log_path = /var/log/mmm/audit.log
audit_retention_days = 90

```

3. Service Configuration

Create systemd service `/etc/systemd/system/mmm.service` :

```

[Unit]
Description=Master Memory Manager
After=network.target
Requires=network.target

[Service]
Type=forking
User=mmm
Group=mmm
ExecStart=/usr/local/bin/mmm-daemon --config /etc/mmm/mmm.conf
ExecReload=/bin/kill -HUP $MAINPID
ExecStop=/bin/kill -TERM $MAINPID
Restart=always
RestartSec=5
LimitNOFILE=65536
LimitMEMLOCK=infinity

[Install]
WantedBy=multi-user.target

```

Deployment Steps

1. User and Permissions Setup

```

# Create MMM user and group
sudo groupadd mmm
sudo useradd -r -g mmm -s /bin/false mmm

# Create directories
sudo mkdir -p /etc/mmm /var/log/mmm /var/lib/mmm
sudo chown -R mmm:mmm /var/log/mmm /var/lib/mmm
sudo chmod 755 /etc/mmm
sudo chmod 750 /var/log/mmm /var/lib/mmm

```

2. Install Configuration Files

```

# Copy configuration files
sudo cp config/mmm.conf /etc/mmm/
sudo cp config/security.policy /etc/mmm/
sudo chown root:mmm /etc/mmm/*
sudo chmod 640 /etc/mmm/*

```

3. Enable and Start Service

```

# Reload systemd
sudo systemctl daemon-reload

# Enable MMM service
sudo systemctl enable mmm

# Start MMM service
sudo systemctl start mmm

# Check status
sudo systemctl status mmm

```

4. Verify Deployment

```
# Check service status
sudo systemctl is-active mmm

# Check logs
sudo journalctl -u mmm -f

# Test API endpoints
curl http://localhost:8080/api/v1/status

# Run diagnostic tests
mmm-diagnostic --full-test
```

Monitoring Setup

1. Log Configuration

Configure log rotation in `/etc/logrotate.d/mmm` :

```
/var/log/mmm/*.log {
    daily
    rotate 30
    compress
    delaycompress
    missingok
    notifempty
    create 640 mmm mmm
    postrotate
        systemctl reload mmm
    endscript
}
```

2. Performance Monitoring

```
# Install monitoring tools
sudo apt-get install -y prometheus node-exporter grafana

# Configure Prometheus to scrape MMM metrics
# Add to /etc/prometheus/prometheus.yml:
# - job_name: 'mmm'
#   static_configs:
#     - targets: ['localhost:8080']
```

3. Alerting Setup

Configure alerts in `/etc/mmm/alerts.conf` :

```

alerts:
- name: high_memory_usage
  condition: memory_usage > 90
  action: email,sms
  recipients: admin@company.com

- name: performance_degradation
  condition: response_time > 100ms
  action: auto_optimize

- name: security_violation
  condition: unauthorized_access
  action: block_ip,email
  recipients: security@company.com

```

Performance Tuning

1. Kernel Parameters

Add to `/etc/sysctl.conf` :

```

# MMM performance tuning
vm.swappiness = 1
vm.dirty_ratio = 15
vm.dirty_background_ratio = 5
kernel.shmmax = 68719476736
kernel.shmall = 4294967296

```

2. CPU Affinity

```

# Set CPU affinity for MMM processes
echo "2-7" > /sys/fs/cgroup/cpuset/mmm/cpuset.cpus
echo "1" > /sys/fs/cgroup/cpuset/mmm/cpuset.cpu_exclusive

```

3. Memory Configuration

```

# Configure huge pages
echo 1024 > /proc/sys/vm/nr_hugepages
echo always > /sys/kernel/mm/transparent_hugepage/enabled

```

Troubleshooting

Common Issues

1. Service Won't Start

```
# Check configuration syntax
mmm-config-check /etc/mmm/mmm.conf

# Check permissions
ls -la /etc/mmm/
ls -la /var/log/mmm/

# Check dependencies
ldd /usr/local/bin/mmm-daemon
```

2. Performance Issues

```
# Check system resources
htop
iotop
perf top

# Check MMM metrics
mmm-stats --detailed
curl http://localhost:8080/api/v1/metrics
```

3. Memory Leaks

```
# Enable memory debugging
export MMM_DEBUG_MEMORY=1
systemctl restart mmm

# Monitor memory usage
watch -n 1 'cat /proc/$(pgrep mmm-daemon)/status | grep VmRSS'
```

Log Analysis

```
# Check error logs
sudo grep ERROR /var/log/mmm/mmm.log

# Check performance logs
sudo grep PERFORMANCE /var/log/mmm/performance.log

# Check security logs
sudo grep SECURITY /var/log/mmm/audit.log
```

Backup and Recovery

1. Configuration Backup

```
# Create backup script
#!/bin/bash
BACKUP_DIR="/backup/mmm/$(date +%Y%m%d)"
mkdir -p $BACKUP_DIR
cp -r /etc/mmm/ $BACKUP_DIR/
cp -r /var/lib/mmm/ $BACKUP_DIR/
tar -czf $BACKUP_DIR.tar.gz $BACKUP_DIR/
```

2. Recovery Procedure

```
# Stop service
sudo systemctl stop mmm

# Restore configuration
sudo tar -xzf /backup/mmm/20231201.tar.gz -C /

# Fix permissions
sudo chown -R mmm:mmm /var/lib/mmm/
sudo chown root:mmm /etc/mmm/*

# Start service
sudo systemctl start mmm
```

Security Hardening

1. Network Security

```
# Configure firewall
sudo ufw allow from 10.0.0.0/8 to any port 8080
sudo ufw deny 8080
```

2. File Permissions

```
# Secure configuration files
sudo chmod 600 /etc/mmm/security.policy
sudo chmod 640 /etc/mmm/mmm.conf
```

3. SELinux/AppArmor

Configure appropriate security policies for your security framework.

Scaling and High Availability

1. Load Balancing

Configure multiple MMM instances behind a load balancer for high availability.

2. Clustering

Use MMM's built-in clustering capabilities for distributed deployments.

3. Auto-Scaling

Configure auto-scaling based on system metrics and load patterns.

Support and Maintenance

Regular Maintenance Tasks

- Monitor system performance and logs daily
- Update configuration as needed
- Perform regular backups
- Apply security updates promptly
- Review and rotate logs monthly

Getting Help

- Check documentation: `/usr/local/share/doc/mmm/`
- Run diagnostics: `mmm-diagnostic --help`
- Community support: GitHub Issues
- Enterprise support: Contact Mecca Research

This deployment guide ensures a robust, secure, and performant MMM installation ready for production workloads.